TGA-MGNet-Lite: Supplementary Material

Param Shah
Ahmedabad University
India
param.s4@ahduni.edu.in

Mehul S. Raval
Ahmedabad University
India
mehul.raval@ahduni.edu.in

Dhruv Dhirendra Premani
Ahmedabad University
India
dhruv.p10@ahduni.edu.in

ACM Reference Format:
Param Shah, Mehul S. Raval, and Dhruv Dhirendra Premani. 2026. TGA-MGNet-Lite: Supplementary Material. In *Indian Conference on Computer Vision, Graphics, and Image Processing (ICVGIP 2026), December 21–24, 2026, Kolkata, India.* ACM, New York, NY, USA, 2 pages. https://doi.org/xx.xxxx/xxxxxxx.xxxxxxx

This supplement contains material moved out of the main manuscript for length. Section and table references (e.g., Section 3.1, Table 5) refer to the main paper.

S1. Per-Zone Gland Segmentation Dice

Since the same six zones used for AMCP's motivating scale variation (thick central vs. thin temporal glands) also feed the grading transformer as regional tokens, we report gland Dice broken down by zone (Table S1). Central zones Z_2–Z_4 reach 0.81–0.83 Dice, while the temporal extremities Z_1 and Z_6 trail at 0.70–0.71, a ∼0.12 gap consistent with thinner, more fragmented gland tissue near the eyelid margins being harder to segment. This indicates AMCP narrows but does not eliminate the scale-variation challenge it targets, and it is a relevant caveat since noisier zone masks at Z_1/Z_6 propagate directly into the corresponding regional tokens used for grading (referenced in the main text's Model Complexity and Segmentation and Component-Level Ablation Study sections as the plausible mechanism behind zone tokens' measured effect).

Table S1: Per-Zone Gland Segmentation Dice (Test Set, N=153)

Zone	Mean Dice	Std	N
Global	0.7955	0.0532	153
Z_1	0.7024	0.1414	153
Z_2	0.8087	0.0818	153
Z_3	0.8311	0.0712	153
Z_4	0.8236	0.0834	153
Z_5	0.8052	0.0999	153
Z_6	0.7064	0.1292	153

S2. CORAL-Style Threshold Sensitivity Analysis

Validation-tuned CORAL-style thresholds are required to reproduce this model's reported headline performance; literal-default and argmax decoding both perform substantially worse and are not viable alternatives for this model. The validation-tuned thresholds (0.25, 0.60, 0.60) reproduce the model's headline test performance (QWK 0.7567); the literal (0.50, 0.50, 0.50) decode instead gives QWK 0.6313, and naive argmax decoding (which by construction applies no calibration at all) collapses further, to QWK 0.5753 with 4 non-adjacent errors, its worst ordinal-consistency result in this work.

Table S2: CORAL-Style Threshold Sensitivity Analysis (Test Set)

Threshold Set	Acc	QWK	F1	N-Adj
Literal default (.50, .50, .50)	0.4444	0.6313	0.5232	1
Validation-tuned (.25, .60, .60)	0.7516	0.7567	0.6634	0
Perturbed −0.05	0.7320	0.7354	0.6652	0
Perturbed +0.05	0.7451	0.7537	0.6550	0
Perturbed −0.10	0.6732	0.6575	0.5770	0
Perturbed +0.10	0.7124	0.7447	0.6718	0
Argmax (reconstructed class probs.)†	0.3399	0.5753	0.4284	4

† Argmax over class probabilities reconstructed from the cumulative CORAL-style outputs via Eq. (2) below. Majority-class baseline for this test set: 60.78% accuracy (93/153).

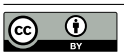
Note on the argmax row. The CORAL-style head (independent per-threshold weights, as described in the main paper) emits $K-1$ *cumulative* logits $P(y{>}k)$ rather than K class scores, so an argmax decode requires first converting these to class probabilities. The literal-default row above is the rank decode, which thresholds the cumulative probabilities directly:

$$\hat{y} \;=\; \sum_{k=0}^{K-2} \mathbb{I}[P(y{>}k) > 0.5]\,, \tag{1}$$

where $\mathbb{I}[\cdot]$ is the indicator function and K=4. The Argmax row is the second standard alternative: an argmax over class probabilities reconstructed from the cumulative outputs,

$$\begin{aligned} p_0 &= 1 - P(y{>}0), \\ p_j &= P(y{>}j{-}1) - P(y{>}j), \quad j = 1, \ldots, K-2, \\ p_{K-1} &= P(y{>}K{-}2), \end{aligned} \tag{2}$$

with p_j clipped to $[0, \infty)$ and renormalized to sum to 1 before the argmax, to handle the non-monotonic cumulative outputs that an uncalibrated CORAL-style head can produce; giving $\hat{y} = \arg\max_j p_j$. Both rows in Table S2 are therefore standard decodes, computed directly from the model's stored test-set probabilities. The Argmax row's accuracy (33.99%) falls below the 60.78% majority-class baseline despite the model's ROC-AUC of 0.9051, which reflects the CORAL head's uncalibrated cumulative outputs rather than a non-standard decoding choice, and underscores why validation-tuned thresholds (rather than either standard untuned decode) are required to reproduce the headline performance.



This work is licensed under a Creative Commons Attribution 4.0 International License.
ICVGIP 2026, Kolkata, India
© 2026 Copyright held by the owner/author(s).
ACM ISBN xxx-x-xxxx-xxxx-x/2026/12
https://doi.org/xx.xxxx/xxxxxxx.xxxxxxx

Note on this row's numerical values. The evaluation code underlying this row has always reconstructed class probabilities via Eq. (2) (with the clipping and renormalization described above) before taking the argmax; no new computation was run between the earlier and current versions of this supplement. The only change was to this row's label and surrounding prose, which previously and incorrectly described an already-correct Eq. (2) computation as a non-standard decode. The reported values (0.3399, 0.5753, 0.4284, 4) are therefore identical across both versions by construction, not by coincidence.

Small perturbations to the tuned thresholds (between ±0.05 and ±0.10) degrade QWK somewhat but stay well above literal-default or argmax decoding, indicating the model's CORAL-style output is not well-calibrated at the nominal 0.5 cumulative-probability boundary and genuinely requires threshold calibration on held-out data. We note that this calibration was tuned on only 149 validation samples and could still be somewhat optimistic, but we have no evidence that literal defaults offer a safer or better-generalizing alternative on this test set.